

Direct Discovery of communities in a social graph

By

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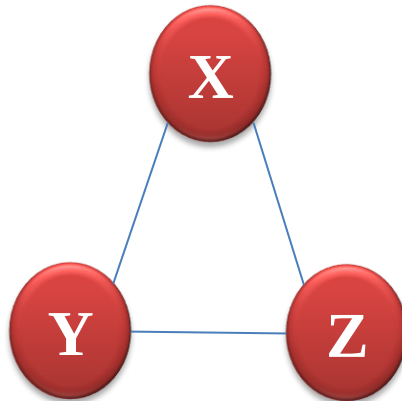
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Clique Percolation Method (CPM)

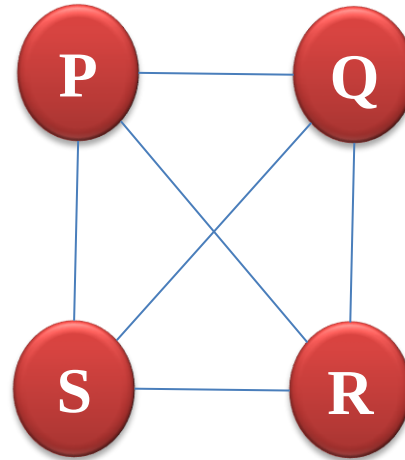
- Clique is group of nodes in graph such that all nodes in a clique are connected to each other.
- 'k' $\Rightarrow$ No. of nodes in a clique



For k = 2



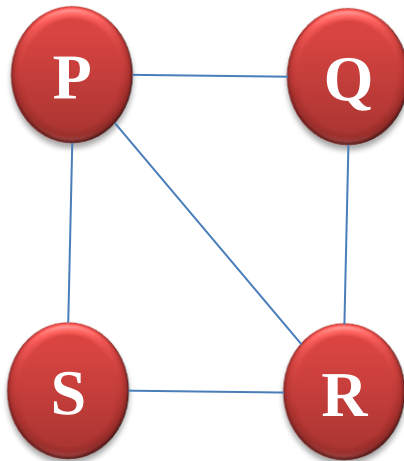
For k = 3



For k = 4

Community

- Community is the group of cliques such that all cliques must have 'k-1' nodes in common.



Here, for $k=3$

Clique1 = {P, Q, R}

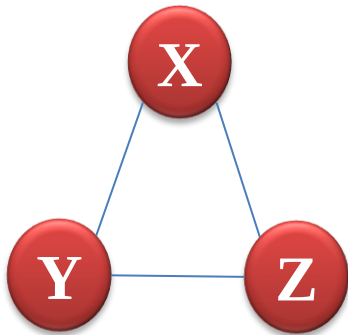
Clique2 = {P, R, S}

Community = {Clique1, Clique2}

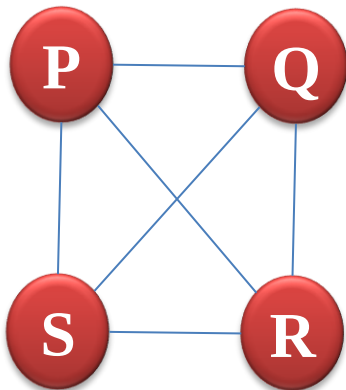
Diagram for Community



Community = {A,B}



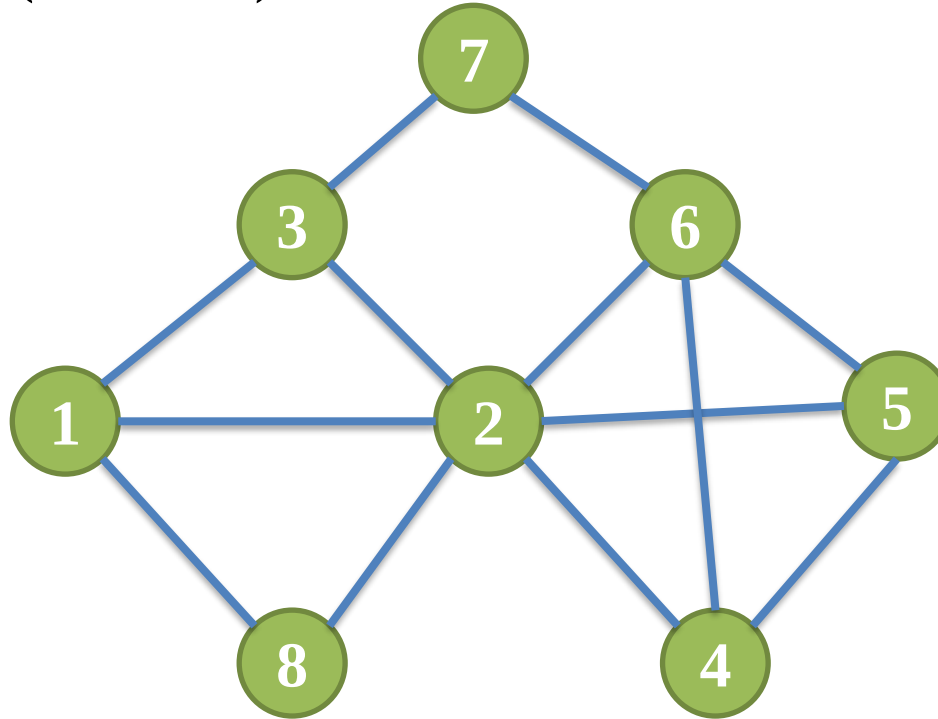
Community = {X, Y, Z}

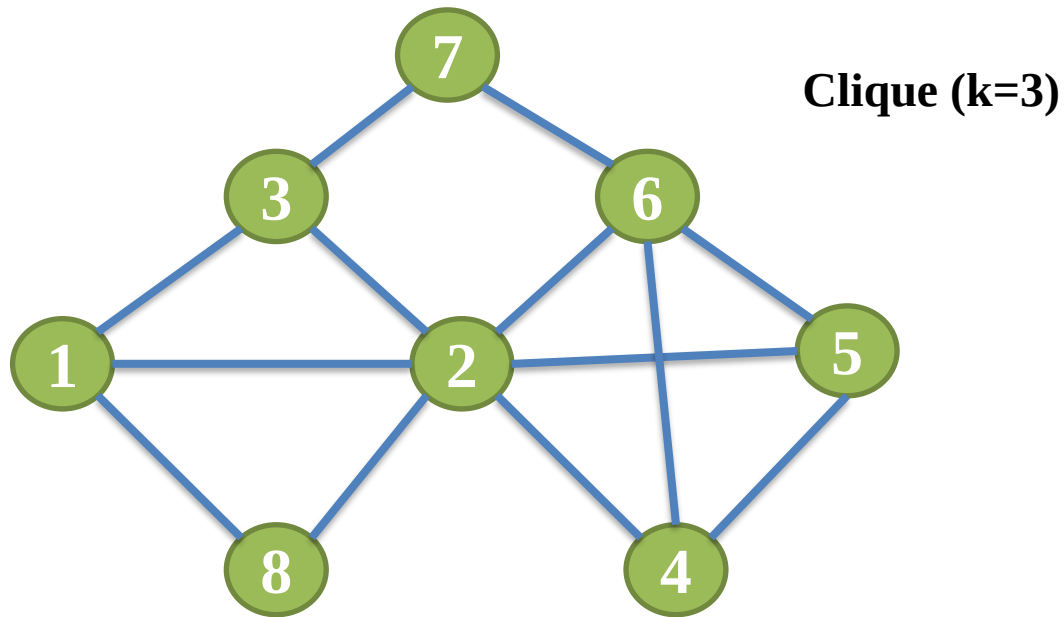


Community = {P,Q,R,S}

Example

Use clique percolation method (CPM) and find the communities ?(For $k=3$)





- {1,2,3}
- {1,2,8}
- {2,6,5}
- {2,6,4}
- {2,5,4}
- {4,5,6}

Thank You